THE BITCOIN STANDARD

The Decentralized Alternative to Central Banking

Saifedean Ammous

WILEY

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To my wife and daughter, who give me a reason to write.
And to Satoshi Nakamoto, who gave me something worth writing about.

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About the Author

Saifedean Ammous is an independent scholar researching and teaching bitcoin and economics in the Austrian school tradition on his website: www.saifedean.com.

Author's preface to the 2021 updated printing

Three years after the initial publication of *The Bitcoin Standard* in 2018, interest in the book has continued to grow steadily, with each of the first three months of 2021 registering a record in monthly sales, and translations rights sold for 25 languages. The growing demand made me consider publishing a second edition of the book, correcting some unfortunate errors, updating the book with the data and developments of the intervening three years, and adding two new chapters to discuss bitcoin's scaling and energy consumption in more depth. But a lot has changed in the world of bitcoin over the last three years, and introducing the data and developments of these years would require significant edits to the text. *The Bitcoin Standard*, and its analysis and conclusions, was the product of its time, a snapshot of a specific point in bitcoin's history, after the contentious hard fork wars were decisively resolved in a way that cemented bitcoin's value proposition of an immutable hard monetary policy, when the total value of the bitcoin network had settled around the milestone of $100 billion, just before the network was to enter its second decade and mature into an

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investment-grade asset class attracting the attention of the world's major financial institutions and corporations. While the events of the last three years do not invalidate the analysis and conclusions of the book, including them would change the book significantly from the original text, popular with readers around the world, and result in the creation of a different book, a product of a different time in bitcoin's life.

Instead of a significantly altered second edition, and in consultation with my editor, I decided to let The Bitcoin Standard live on in its original form, with only some minor adjustments to correct a few obvious mistakes, some of which were glaring, though inconsequential to the substance and arguments of the book. I will instead include new developments and analysis in a sequel, *The Fiat Standard: The Debt Slavery Alternative to Human Civilization*. The sequel will analyze the rise of bitcoin by analyzing the workings of the fiat monetary system using the same approach, methods, and terminology used in studying bitcoin in *The Bitcoin Standard*.

Unexpected to me at the time of writing, *The Bitcoin Standard* has proven popular among corporations and financial institutions entering the bitcoin space. The most astonishing compliment the book received was when Michael Saylor, CEO of publicly traded software intelligence firm Microstrategy, decided to put his company on a bitcoin standard, making bitcoin its prime treasury reserve asset. Mr. Saylor has not only backed his conviction in bitcoin by putting significant amounts of his personal wealth and company treasury in bitcoin, but he has, in a very short period of time, become one of the leading thinkers and analysts of bitcoin, as well as one of its most effective communicators. The conviction, courage, and clarity of Mr. Saylor in pursuing a bitcoin standard has been an inspiration, and I am honored he has agreed to write a foreword to include with this reprint, to contextualize the importance of this book for the continued emergence of a global bitcoin standard.

With the revisions complete, and a new foreword to introduce the book from the world's pioneering leader in adopting the bitcoin standard, I hope readers find this version of the book worthy of a place on their bookshelves, and in their recommendations to family, friends, and colleagues, well into the future.

Saifedean Ammous
Amman, Jordan
April 27, 2021



Foreword

by Michael J. Saylor

In March 2020, the world was struck by a pandemic, which brought entire segments of our economy to a grinding halt, and interrupted patterns of behavior that we had all grown accustomed to over the course of hundreds of years. Offices and schools closed, public gatherings ceased, commercial and recreational travel were stopped, and in-person meetings became impossible or impractical. Stores were closed, factories idled, aircraft were grounded, ships anchored, roads were blocked, and borders closed.

The monetary response of policymakers to this economic shock was unprecedented. The supply of currency expanded at the fastest rate in modern history, with every nation engaged in aggressive asset purchases, fiscal stimulus, deficit spending, and interest rate suppression. The result was a K-shaped recovery—asset-rich firms rapidly bounced back and had their best performing year of the century, while operating companies saw their revenues collapse and their earnings dissipate.

The cognitive dissonance was bewildering, as was the massive wealth redistribution. Purely digital firms saw their demand and revenues explode upwards. Brick-and-mortar retail, travel, hospitality,

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and entertainment industries struggled to stay solvent. MicroStrategy was positioned in the middle of this economic landscape—and we spent the entire second quarter transforming our operations to become digital first, by rebuilding the sales, marketing, and services operations around our website, and implementing videoconferencing, automation, cloud services, and remote work.

By June, we had done enough work and gained enough information to conclude that the five-hundred-million-dollar war chest that we were saving for a rainy day would be of no use in the new virtual world, and we would probably generate an additional five hundred million in cash flow due to our digital transformation. The good news was that we had plenty of cash and a high prospect of generating more over time. The bad news was that the monetary inflation rate had tripled, and the price of other assets was exploding at the fastest rate in a generation. Our treasury was a rapidly melting ice cube, and we needed to act quickly if we didn't want to see all the value it represented waste away.

This catalyzed a mad scramble for a solution to our problem. How does a modern corporation protect its balance sheet in an environment of monetary inflation where the currency is losing 15% of its purchasing power each year, while the after-tax yields available from traditional treasury instruments are effectively zero? A company generating seventy-five million per year in cash flow, holding five hundred million dollars in treasury balances at a negative real yield of 15% is destroying as much shareholder value as it is creating. In essence, we were running as hard as we could to stand still.

After considering and dismissing cash, bonds, real estate, equity, derivatives, art, commodities, and collectibles as treasury assets, we were left with just precious metals and cryptocurrencies. It was at this point in my search for a solution that I discovered *The Bitcoin Standard* by Saifedean Ammous, and it was this book, more than any other, that provided the holistic economic framework that I needed to interpret the macroeconomic forces reshaping our world, distorting our markets, and buffeting corporations.

The Bitcoin Standard should be required reading for everyone in modern society. It offers a concise, coherent narrative of monetary theory, the history of money, practical economics, and the impact of political policy on business, culture, and the economy. The book contains perhaps one

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of the best articulations of the virtues of strong money and the dangers of weak currency yet presented in modern business literature. *The Bitcoin Standard* also masterfully debunks the myths of modern monetary theory and the broken ideas that have dominated the fiat economic school of thought since the early twentieth century.

In May 2020, this book was critical in leading me to conclude that bitcoin was the solution to our corporate treasury problem. Our firm elected to invest our cash assets in bitcoin in August 2020, eventually adopting it as our primary treasury reserve asset and purchasing $2.2 billion in BTC over the following six months. *The Bitcoin Standard* helped us realize that the best business strategy for our firm was to hold just a small working capital balance in fiat currencies, sweeping the rest of our cash flows into our treasury and converting those sums into bitcoin as soon as practical. As I write these words, 99% of our assets are stored in bitcoin, with the remaining 1% stored in the local currencies required to do business in various markets. In essence, MicroStrategy has adopted the bitcoin standard.

The Bitcoin Standard is my first recommendation for those seeking a holistic appreciation of the economic theory, political history, and technological developments that have driven the growth of the bitcoin network and define its future trajectory. It is suitable for individuals, investors, executives, technologists, politicians, journalists, and academics alike, regardless of their agenda. Bitcoin is the world's first digital monetary network. Bitcoin is also the world's first engineered monetary asset. Together, these traits represent the most disruptive technology in the world, the greatest opportunity currently available to those who wish to create something new and wonderful, and the solution to the store-of-value problem faced by 7.8 billion people, 100+ million companies, and hundreds of trillions of dollars of investor capital.

I hope you enjoy this book as much as I did, and benefit from ideas contained within these pages.

Michael J. Saylor
Chairman & CEO
MicroStrategy
Miami Beach, FL
24 March 2021

Prologue

On October 31, 2008, a computer programmer going by the pseudonym Satoshi Nakamoto sent an email to a cryptography mailing list to announce that he had produced a "new electronic cash system that's fully peer-to-peer, with no trusted third party."[1] He copied the abstract of the paper explaining the design, and a link to it online. In essence, bitcoin offered a payment network with its own native currency, and used a sophisticated method for members to verify all transactions without having to trust in any single member of the network. The currency was issued at a predetermined rate to reward the members who spent their processing power on verifying the transactions, thus providing a reward for their work. The startling thing about this invention was that, contrary to many other previous attempts at setting up a digital cash, it actually worked.

While a clever and neat design, there wasn't much to suggest that such a quirky experiment would interest anyone outside the circles of cryptography geeks. For months this was the case, as barely a few dozen users worldwide were joining the network and engaging in mining and

[1] The full email can be found on the Satoshi Nakamoto Institute archive of all known Satoshi Nakamoto writings, available at www.nakamotoinstitute.org. "Re: Bitcoin P2P e-cash paper." Received by The Cryptography Mailing List, 31 Oct. 2008.

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sending each other coins that began to acquire the status of collectibles, albeit in digital form.

But in October 2009, an Internet exchange[2] sold 5,050 bitcoins for $5.02, at a price of $1 for 1,006 bitcoins, to register the first purchase of a bitcoin with money.[3] The price was calculated by measuring the value of the electricity needed to produce a bitcoin. In economic terms, this seminal moment was arguably the most significant in bitcoin's life. Bitcoin was no longer just a digital game being played within a fringe community of programmers; it had now become a market good with a price, indicating that someone somewhere had developed a positive valuation for it. On May 22, 2010, someone else paid 10,000 bitcoins to buy two pizza pies worth $25, representing the first time that bitcoin was used as a medium of exchange. The token had needed seven months to transition from being a market good to being a medium of exchange.

Since then, the bitcoin network has grown in the number of users and transactions, and the processing power dedicated to it, while the value of its currency has risen quickly, exceeding $7,000 per bitcoin as of November 2017.[4] After eight years, it is clear that this invention is no longer just an online game, but a technology that has passed the market test and is being used by many for real-world purposes, with its exchange rate being regularly featured on TV, in newspapers, and on websites along with the exchange rates of national currencies.

Bitcoin can be best understood as distributed software that allows for transfer of value using a currency protected from unexpected inflation without relying on trusted third parties. In other words, bitcoin automates the functions of a modern central bank and makes them predictable and virtually immutable by programming them into code decentralized among thousands of network members, none of whom can alter the code without the consent of the rest. This makes bitcoin the first demonstrably reliable operational example of *digital cash* and *digital hard money*. While bitcoin is a new invention of the digital age, the problems it purports to solve—namely, providing a form of money that is

[2]The now-defunct *New Liberty Standard*, newlibertystandard.wikifoundry.com/.

[3]Popper, Nathaniel. *Digital Gold: Bitcoin and the Inside Story of the Misfits and Millionaires Trying to Reinvent Money*. HarperCollins, 2015.

[4]In other words, in the eight years it has been a market commodity, a bitcoin has appreciated around almost eight million-fold, or, precisely 793,513,944% from its first price of $0.000994 to its all-time high at the time of writing, $7,888.

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under the full command of its owner and likely to hold its value in the long run—are as old as human society itself. This book presents a conception of these problems based on years of studying this technology and the economic problems it solves, and how societies have previously found solutions for them throughout history. My conclusion may surprise those who label bitcoin a scam or ruse of speculators and promoters out to make a quick buck. Indeed, bitcoin improves on earlier "store of value" solutions, and bitcoin's suitability as the sound money of a digital age may catch naysayers by surprise.

History can foreshadow what's to come, particularly when examined closely. And time will tell just how sound the case made in this book is. As it must, the first part of the book explains money, its function and properties. As an economist with an engineering background, I have always sought to understand a technology in terms of the problems it purports to solve, which allows for the identification of its functional essence and its separation from incidental, cosmetic, and insignificant characteristics. By understanding the problems money attempts to solve, it becomes possible to elucidate what makes for sound and unsound money, and to apply that conceptual framework to understand how and why various goods, such as seashells, beads, metals, and government money, have served the function of money, and how and why they may have failed at it or served society's purposes to store value and exchange it.

The second part of the book discusses the individual, social, and global implications of sound and unsound forms of money throughout history. Sound money allows people to think about the long term and to save and invest more for the future. Saving and investing for the long run are the key to capital accumulation and the advance of human civilization. Money is the information and measurement system of an economy, and sound money is what allows trade, investment, and entrepreneurship to proceed on a solid basis, whereas unsound money throws these processes into disarray. Sound money is also an essential element of a free society as it provides for an effective bulwark against despotic government.

The third section of the book explains the operation of the bitcoin network and its most salient economic characteristics, and analyzes the possible uses of bitcoin as a form of sound money, discussing some use

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cases which bitcoin does not serve well, as well as addressing some of the most common misunderstandings and misconceptions surrounding it.

This book is written to help the reader understand the economics of bitcoin and how it serves as the digital iteration of the many technologies used to fulfill the functions of money throughout history. This book is not an advertisement or invitation to buy into the bitcoin currency. Far from it. The value of bitcoin is likely to remain volatile, at least for a while; the bitcoin network may yet succeed or fail, for whatever foreseeable or unforeseeable reasons; and using it requires technical competence and carries risks that make it unsuited for many people. This book does not offer investment advice, but aims at helping elucidate the economic properties of the network and its operation, to provide readers an informed understanding of bitcoin before deciding whether they want to use it.

Only with such an understanding, and only after extensive and thorough research into the practical operational aspects of owning and storing bitcoins, should anyone consider holding value in bitcoin. While bitcoin's rise in market value may make it appear like a no-brainer as an investment, a closer look at the myriad hacks, attacks, scams, and security failures that have cost people their bitcoins provides a sobering warning to anyone who thinks that owning bitcoins provides a guaranteed profit. Should you come out of reading this book thinking that the bitcoin currency is something worth owning, your first investment should not be in buying bitcoins, but in time spent understanding how to buy, store, and own bitcoins securely. It is the inherent nature of bitcoin that such knowledge cannot be delegated or outsourced. There is no alternative to personal responsibility for anyone interested in using this network, and that is the real investment that needs to be made to get into bitcoin.

Chapter 1

Money

Bitcoin is the newest technology to serve the function of money—an invention leveraging the technological possibilities of the digital age to solve a problem that has persisted for all of humanity's existence: how to move economic value across time and space. In order to understand bitcoin, one must first understand money, and to understand money, there is no alternative to the study of the function and history of money.

The simplest way for people to exchange value is to exchange valuable goods with one another. This process of *direct exchange* is referred to as barter, but is only practical in small circles with only a few goods and services produced. In a hypothetical economy of a dozen people isolated from the world, there is not much scope for specialization and trade, and it would be possible for individuals to each engage in the production of the most basic essentials of survival and exchange them among themselves directly. Barter has always existed in human society and continues to this day, but it is highly impractical and remains only in

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use in exceptional circumstances, usually involving people with extensive familiarity with one another.

In a more sophisticated and larger economy, the opportunity arises for individuals to specialize in the production of more goods and to exchange them with many more people—people with whom they have no personal relationships, strangers with whom it is utterly impractical to keep a running tally of goods, services, and favors. The larger the market, the more the opportunities for specialization and exchange, but also the bigger the problem of *coincidence of wants*—what you want to acquire is produced by someone who doesn't want what you have to sell. The problem is deeper than different requirements for different goods, as there are three distinct dimensions to the problem.

First, there is the lack of coincidence in scales: what you want may not be equal in value to what you have and dividing one of them into smaller units may not be practical. Imagine wanting to sell shoes for a house; you cannot buy the house in small pieces each equivalent in value to a pair of shoes, nor does the homeowner want to own all the shoes whose value is equivalent to that of the house. Second, there is the lack of coincidence in time frames: what you want to sell may be perishable but what you want to buy is more durable and valuable, making it hard to accumulate enough of your perishable good to exchange for the durable good at one point in time. It is not easy to accumulate enough apples to be exchanged for a car at once, because they will rot before the deal can be completed. Third, there is the lack of coincidence of locations: you may want to sell a house in one place to buy a house in another location, and (most) houses aren't transportable. These three problems make direct exchange highly impractical and result in people needing to resort to performing more layers of exchange to satisfy their economic needs.

The only way around this is through *indirect exchange*: you try to find some other good that another person would want and find someone who will exchange it with you for what you want to sell. That intermediary good is a *medium of exchange*, and while any good could serve as the medium of exchange, as the scope and size of the economy grows it becomes impractical for people to constantly search for different goods that their counterparty is looking for, carrying out several exchanges for each exchange they want to conduct. A far more efficient solution will

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naturally emerge, if only because those who chance upon it will be far more productive than those who do not: a single medium of exchange (or at most a small number of media of exchange) emerges for everyone to trade their goods for. A good that assumes the role of a widely accepted medium of exchange is called money.

Being a medium of exchange is the quintessential function that defines money—in other words, it is a good purchased not to be consumed (a consumption good), nor to be employed in the production of other goods (an investment, or capital good), but primarily for the sake of being exchanged for other goods. While investment is also meant to produce income to be exchanged for other goods, it is distinct from money in three respects: first, it offers a return, which money does not offer; second, it always involves a risk of failure, whereas money is supposed to carry the least risk; third, investments are less liquid than money, necessitating significant transaction costs every time they are to be spent. This can help us understand why there will always be demand for money, and why holding investments can never entirely replace money. Human life is lived with uncertainty as a given, and humans cannot know for sure when they will need what amount of money.[1] It is common sense, and age-old wisdom in virtually all human cultures, for individuals to want to store some portion of their wealth in the form of money, because it is the most liquid holding possible, allowing the holder to quickly liquidate if she needs to, and because it involves less risk than any investment. The price for the convenience of holding money comes in the form of the forgone consumption that could have been had with it, and in the form of the forgone returns that could have been made from investing it.

From examining such human choices in market situations, Carl Menger, the father of the Austrian school of economics and founder of marginal analysis in economics, came up with an understanding of the key property that leads to a good being adopted freely as money on the market, and that is *salability*—the ease with which a good can be

[1]See Ludwig von Mises' *Human Action,* p. 250, for a discussion of how uncertainty about the future is the key driver of demand for holding money. With no uncertainty of the future, humans could know all their incomes and expenditures ahead of time and plan them optimally so they never have to hold any cash. But as uncertainty is an inevitable part of life, people must continue to hold money so they have the ability to spend without having to know the future.

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sold on the market whenever its holder desires, with the least loss in its price.[2]

There is nothing in principle that stipulates what should or should not be used as money. Any person choosing to purchase something not for its own sake, but with the aim of exchanging it for something else, is making it de facto money, and as people vary, so do their opinions on, and choices of, what constitutes money. Throughout human history, many things have served the function of money: gold and silver, most notably, but also copper, seashells, large stones, salt, cattle, government paper, precious stones, and even alcohol and cigarettes in certain conditions. People's choices are subjective, and so there is no "right" and "wrong" choice of money. There are, however, consequences to choices.

The relative salability of goods can be assessed in terms of how well they address the three facets of the problem of the lack of coincidence of wants mentioned earlier: their salability across scales, across space, and across time. A good that is salable across scales can be conveniently divided into smaller units or grouped into larger units, thus allowing the holder to sell it in whichever quantity he desires. Salability across space indicates an ease of transporting the good or carrying it along as a person travels, and this has led to good monetary media generally having high value per unit of weight. Both of these characteristics are not very hard to fulfill by a large number of goods that could potentially serve the function of money. It is the third element, salability across time, which is the most crucial.

A good's salability across time refers to its ability to hold value into the future, allowing the holder to store wealth in it, which is the second function of money: *store of value*. For a good to be salable across time it has to be immune to rot, corrosion, and other types of deterioration. It is safe to say anyone who thought he could store his wealth for the long term in fish, apples, or oranges learned this lesson the hard way, and likely had very little reason to worry about storing wealth for a while. Physical integrity through time, however, is a necessary but insufficient condition for salability across time, as it is possible for a good to lose its value significantly even if its physical condition remains unchanged.

[2]Menger, Carl. "On the Origins of Money." Trans. C. A. Foley. *Economic Journal*, vol. 2, 1892, pp. 239–55.

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For the good to maintain its value, it is also necessary that the supply of the good not increase too drastically during the period in which the holder owns it. A common characteristic of forms of money throughout history is the presence of some mechanism to restrain the production of new units of the good to maintain the value of the existing units. The relative difficulty of producing new monetary units determines the hardness of money: money whose supply is hard to increase is known as *hard money*, while *easy money* is money whose supply is amenable to large increases.

We can understand money's hardness through understanding two distinct quantities related to the supply of a good: (1) the *stock*, which is its existing supply, consisting of everything that has been produced in the past, minus everything that has been consumed or destroyed; and (2) the *flow*, which is the extra production that will be made in the next time period. The ratio between the stock and flow is a reliable indicator of a good's hardness as money, and how well it is suited to playing a monetary role. A good that has a low ratio of stock-to-flow is one whose existing supply can be increased drastically if people start using it as a store of value. Such a good would be unlikely to maintain value if chosen as a store of value. The higher the ratio of the stock to the flow, the more likely a good is to maintain its value over time and thus be more salable across time.[3]

If people choose a hard money, with a high stock-to-flow ratio, as a store of value, their purchasing of it to store it would increase demand for it, causing a rise in its price, which would incentivize its producers to make more of it. But because the flow is small compared to the existing supply, even a large increase in the new production is unlikely to depress the price significantly. On the other hand, if people chose to store their wealth in an easy money, with a low stock-to-flow ratio, it would be trivial for the producers of this good to create very large quantities of it that depress the price, devaluing the good, expropriating the wealth of the savers, and destroying the good's salability across time.

I like to call this the *easy money trap*: anything used as a store of value will have its supply increased, and anything whose supply can be easily

[3]Fekete, Antal. *Whither Gold?* Winner of the 1996 International Currency Prize, Sponsored by Bank Lips, 1997, www.professorfekete.com/articles/AEFWhitherGold.pdf.



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increased will destroy the wealth of those who used it as a store of value. The corollary to this trap is that anything that is successfully used as money will have some natural or artificial mechanism that restricts the new flow of the good into the market, maintaining its value across time. It therefore follows that for something to assume a monetary role, it has to be costly to produce, otherwise the temptation to make money on the cheap will destroy the wealth of the savers, and destroy the incentive anyone has to save in this medium.

Whenever a natural, technological, or political development resulted in quickly increasing the new supply of a monetary good, the good would lose its monetary status and be replaced by other media of exchange with a more reliably high stock-to-flow ratio, as will be discussed in the next chapter. Seashells were used as money when they were hard to find, loose cigarettes are used as money in prisons because they are hard to procure or produce, and with national currencies, the lower the rate of increase of the supply, the more likely the currency is to be held by individuals and maintain its value over time.

When modern technology made the importation and catching of seashells easy, societies that used them switched to metal or paper money, and when a government increases its currency's supply, its citizens shift to holding foreign currencies, gold, or other more reliable monetary assets. The twentieth century provided us an unfortunately enormous number of such tragic examples, particularly from developing countries. The monetary media that survived the longest are the ones that had very reliable mechanisms for restricting their supply growth—in other words, *hard money*. Competition is at all times alive between monetary media, and its outcomes are foretold through the effects of technology on the differing stock-to-flow ratio of the competitors, as will be demonstrated in the next chapter.

While people are generally free to use whichever goods they please as their media of exchange, the reality is that over time, the ones who use hard money will benefit most, by losing very little value due to the negligible new supply of their medium of exchange. Those who choose easy money will likely lose value as its supply grows quickly, bringing its market price down. Whether through prospective rational calculation, or the retrospective harsh lessons of reality, the majority of money and wealth will be concentrated with those who choose the hardest and most

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salable forms of money. But the hardness and salability of goods itself is not something that is static in time. As the technological capabilities of different societies and eras have varied, so has the hardness of various forms of money, and with it their salability. In reality, the choice of what makes the best money has always been determined by the technological realities of societies shaping the salability of different goods. Hence, Austrian economists are rarely dogmatic or objectivist in their definition of sound money. They define it not as a specific good or commodity, but as whichever money emerges on the market, freely chosen by the people who transact with it. It is not imposed by coercive authority, and its value is determined through market interaction, not government imposition.[4] Free-market monetary competition is ruthlessly effective at producing sound money, as it only allows those who choose the right money to maintain considerable wealth over time. There is no need for government to impose the hardest money on society; society will have uncovered it long before it concocted its government, and any governmental imposition, if it were to have any effect, would only serve to hinder the process of monetary competition.

The full individual and societal implications of hard and easy money are far more profound than mere financial loss or gain, and are a central theme of this book, discussed thoroughly in Chapters 5, 6, and 7. Those who are able to save their wealth in a good store of value are likely to plan for the future more than those who have bad stores of value. The soundness of the monetary media, in terms of its ability to hold value over time, is a key determinant of how much individuals value the present over the future, or their *time preference*, a pivotal concept in this book.

Beyond the stock-to-flow ratio, another important aspect of a monetary medium's salability is its acceptability by others. The more people accept a monetary medium, the more liquid it is, and the more likely it is to be bought and sold without too much loss. In social settings with many peer-to-peer interactions, as computing protocols demonstrate, it is natural for a few standards to emerge to dominate exchange, because the gains from joining a network grow exponentially the larger the size of the network. Hence, Facebook and a handful of social media networks

[4]Salerno, Joseph. *Money: Sound and Unsound*. Auburn, AL, Ludwig von Mises Institute, 2010, pp. xiv–xv.

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dominate the market, when many hundreds of almost identical networks were created and promoted. Similarly, any device that sends emails has to utilize the IMAP/POP3 protocol for receiving email, and the SMTP protocol for sending it. Many other protocols were invented, and they could be used perfectly well, but almost nobody uses them because to do so would preclude a user from interacting with almost everyone who uses email today, because they are on IMAP/POP3 and SMTP. Similarly, with money, it was inevitable that one, or a few, goods would emerge as the main medium of exchange, because the property of being exchanged easily matters the most. A medium of exchange, as mentioned before, is not acquired for its own properties, but for its salability.

Further, wide acceptance of a medium of exchange allows all prices to be expressed in its terms, which allows it to play the third function of money: *unit of account*. In an economy with no recognized medium of exchange, each good will have to be priced in terms of each other good, leading to a large number of prices, making economic calculations exceedingly difficult. In an economy with a medium of exchange, all prices of all goods are expressed in terms of the same unit of account. In this society money serves as a metric with which to measure interpersonal value; it rewards producers to the extent that they contribute value to others, and signifies to consumers how much they need to pay to obtain their desired goods. Only with a uniform medium of exchange acting as a unit of account does complex economic calculation become possible, and with it comes the possibility for specialization into complex tasks, capital accumulation, and large markets. The operation of a market economy is dependent on prices, and prices, to be accurate, are dependent on a common medium of exchange, which reflects the relative scarcity of different goods. If this is easy money, the ability of its issuer to constantly increase its quantity will prevent it from accurately reflecting opportunity costs. Every unpredictable change in the quantity of money would distort its role as a measure of interpersonal value and a conduit for economic information.

Having a single medium of exchange allows the size of the economy to grow as large as the number of people willing to use that medium of exchange. The larger the size of the economy, the larger the opportunities for gains from exchange and specialization, and perhaps more significantly, the longer and more sophisticated the structure of production

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can become. Producers can specialize in producing capital goods that will only produce final consumer goods after longer intervals, which allows for more productive and superior products. In the primitive small economy, the structure of production of fish consisted of individuals going to the shore and catching fish with their bare hands, with the entire process taking a few hours from start to finish. As the economy grows, more sophisticated tools and capital goods are utilized, and the production of these tools stretches the duration of the production process significantly while also increasing its productivity. In the modern world, fish are caught with highly sophisticated boats that take years to build and are operated for decades. These boats are able to sail to seas that smaller boats cannot reach and thus produce fish that would otherwise not be available. The boats can brave inclement weather and continue production in very difficult conditions where less capital-intensive boats would be docked uselessly. As capital accumulation has made the process longer, it has become more productive per unit of labor, and it can produce superior products that were never possible for the primitive economy with basic tools and no capital accumulation. None of this would be possible without money playing the roles of medium of exchange to allow specialization; store of value to create future-orientation and incentivize individuals to direct resources to investment instead of consumption; and unit of account to allow economic calculation of profits and losses.

The history of money's evolution has seen various goods play the role of money, with varying degrees of hardness and soundness, depending on the technological capabilities of each era. From seashells to salt, cattle, silver, gold, and gold-backed government money, ending with the current almost universal use of government-provided legal tender, every step of technological advance has allowed us to utilize a new form of money with added benefits, but, as always, new pitfalls. By examining the history of the tools and materials that have been employed in the role of money throughout history, we are able to discern the characteristics that make for good money and the ones that make for bad money. Only with this background in place can we then move on to understand how bitcoin functions and what its role as a monetary medium is.

The next chapter examines the history of obscure artifacts and objects that have been used as money throughout history, from the Rai stones of Yap Island, to seashells in the Americas, glass beads in Africa,

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and cattle and salt in antiquity. Each of these media of exchange served the function of money for a period during which it had one of the best stock-to-flow ratios available to its population, but stopped when it lost that property. Understanding how and why is essential to understanding the future evolution of money and any likely role bitcoin will play. Chapter 3 moves to the analysis of monetary metals and how gold came to be the prime monetary metal in the world during the era of the gold standard at the end of the nineteenth century. Chapter 4 analyzes the move to government money and its track record. After the economic and social implications of different kinds of money are discussed in Chapters 5, 6, and 7, Chapter 8 introduces the invention of bitcoin and its monetary properties.

Chapter 2

Primitive Moneys

Of all the historical forms of money I have come across, the one that most resembles the operation of bitcoin is the ancient system based on Rai stones on Yap Island, today a part of the Federated States of Micronesia. Understanding how the large circular stones carved from limestone functioned as money will help us explain bitcoin's operation in Chapter 8. Understanding the remarkable tale of how the Rai stones lost their monetary role is an object lesson in how money loses its monetary status once it loses its hardness.

The Rai stones that constituted money were of various sizes, rising to large circular disks with a hole in the middle that weighed up to four metric tons. They were not native to Yap, which did not contain any limestone, and all of Yap's stones were brought in from neighboring Palau or Guam. The beauty and rarity of these stones made them desirable and venerable in Yap, but procuring them was very difficult as it involved a laborious process of quarrying and then shipping them with rafts and canoes. Some of these rocks required hundreds of people to transport them, and once they arrived on Yap, they were placed in a prominent

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location where everyone could see them. The owner of the stone could use it as a payment method without it having to move: all that would happen is that the owner would announce to all townsfolk that the stone's ownership has now moved to the recipient. The whole town would recognize the ownership of the stone and the recipient could then use it to make a payment whenever he so pleased. There was effectively no way of stealing the stone because its ownership was known by everybody.

For centuries, and possibly even millennia, this monetary system worked well for the Yapese. While the stones never moved, they had salability across space, as one could use them for payment anywhere on the island. The different sizes of the different stones provided some degree of salability across scales, as did the possibility of paying with fractions of a single stone. The stones' salability across time was assured for centuries by the difficulty and high cost of acquiring new stones, because they didn't exist in Yap and quarrying and shipping them from Palau was not easy. The very high cost of procuring new stones to Yap meant that the existing supply of stones was always far larger than whatever new supply could be produced at a given period of time, making it prudent to accept them as a form of payment. In other words, Rai stones had a very high stock-to-flow ratio, and no matter how desirable they were, it was not easy for anyone to inflate the supply of stones by bringing in new rocks. Or, at least, that was the case until 1871, when an Irish-American captain by the name of David O'Keefe was shipwrecked on the shores of Yap and revived by the locals.[1]

O'Keefe saw a profit opportunity in taking coconuts from the island and selling them to producers of coconut oil, but he had no means to entice the locals to work for him, because they were very content with their lives as they were, in their tropical paradise, and had no use for whatever foreign forms of money he could offer them. But O'Keefe wouldn't take no for an answer; he sailed to Hong Kong, procured a large boat and explosives, took them to Palau, where he used the explosives and modern tools to quarry several large Rai stones, and set sail to Yap to present the stones to the locals as payment for coconuts. Contrary

[1] The story of O'Keefe inspired the writing of a novel named *His Majesty O'Keefe* by Laurence Klingman and Gerald Green in 1952, which was made into a Hollywood blockbuster by the same name starring Burt Lancaster in 1954.

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to what O'Keefe expected, the villagers were not keen on receiving his stones, and the village chief banned his townsfolk from working for the stones, decreeing that O'Keefe's stones were not of value, because they were gathered too easily. Only the stones quarried traditionally, with the sweat and blood of the Yapese, were to be accepted in Yap. Others on the island disagreed, and they did supply O'Keefe with the coconuts he sought. This resulted in conflict on the island, and in time the demise of Rai stones as money. Today, the stones serve a more ceremonial and cultural role on the island and modern government money is the most commonly used monetary medium.

While O'Keefe's story is highly symbolic, he was but the harbinger of the inevitable demise of Rai stones' monetary role with the encroachment of modern industrial civilization on Yap and its inhabitants. As modern tools and industrial capabilities reached the region, it was inevitable that the production of the stones would become far less costly than before. There would be many O'Keefes, local and foreign, able to supply Yap with an ever-larger flow of new stones. With modern technology, the stock-to-flow ratio for Rai stones decreased drastically: it was possible to produce far more of these stones every year, significantly devaluing the island's existing stock. It became increasingly unwise for anyone to use these stones as a store of value, and thus they lost their salability across time, and with it, their function as a medium of exchange.

The details may differ, but the underlying dynamic of a drop in stock-to-flow ratio has been the same for every form of money that has lost its monetary role, up to the collapse of the Venezuelan bolivar taking place as these lines are being written.

A similar story happened with the aggry beads used as money for centuries in western Africa. The history of these beads in western Africa is not entirely clear, with suggestions that they were made from meteorite stones, or passed on from Egyptian and Phoenician traders. What is known is that they were precious in an area where glassmaking technology was expensive and not very common, giving them a high stock-to-flow ratio, making them salable across time. Being small and valuable, these beads were salable across scale, because they could be combined into chains, necklaces, or bracelets; though this was far from ideal, because there were many different kinds of beads rather than one standard unit. They were also salable across space as they were easy

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to move around. In contrast, glass beads were not expensive and had no monetary role in Europe, because the proliferation of glassmaking technology meant that if they were to be utilized as a monetary unit, their producers could flood the market with them—in other words, they had a low stock-to-flow ratio.

When European explorers and traders visited West Africa in the sixteenth century, they noticed the high value given to these beads and so started importing them in mass quantities from Europe. What followed was similar to the story of O'Keefe, but given the tiny size of the beads and the much larger size of the population, it was a slower, more covert process with bigger and more tragic consequences. Slowly but surely, Europeans were able to purchase a lot of the precious resources of Africa for the beads they acquired back home for very little.[2] European incursion into Africa slowly turned beads from hard money to easy money, destroying their salability and causing the erosion of the purchasing power of these beads over time in the hands of the Africans who owned them, impoverishing them by transferring their wealth to the Europeans, who could acquire the beads easily. The aggry beads later came to be known as slave beads for the role they played in fueling the slave trade of Africans to Europeans and North Americans. A one-time collapse in the value of a monetary medium is tragic, but at least it is over quickly and its holders can begin trading, saving, and calculating with a new one. But a slow drain of its monetary value over time will slowly transfer the wealth of its holders to those who can produce the medium at a low cost. This is a lesson worth remembering when we turn to the discussion of the soundness of government money in the later parts of the book.

Seashells are another monetary medium that was widely used in many places around the world, from North America to Africa and Asia. Historical accounts show that the most salable seashells were usually the ones that were scarcer and harder to find, because these would hold value more than the ones that can be found easily.[3] Native Americans and early European settlers used wampum shells extensively, for the

[2]To maximize their profits, Europeans used to fill the hulls of their boats with large quantities of these beads, which also served to stabilize the boat on its trip.

[3]Szabo, Nick. "Shelling Out: The Origins of Money." *Satoshi Nakamoto Institute*, 2002, nakamotoinstitute.org/shelling-out/.



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same reasons as aggry beads: they were hard to find, giving them a high stock-to-flow ratio, possibly the highest among durable goods available at the time. Seashells also shared with aggry beads the disadvantage of not being uniform units, which meant prices and ratios could not be easily measured and expressed in them uniformly, which creates large obstacles to the growth of the economy and the degree of specialization. European settlers adopted seashells as legal tender from 1636, but as more and more British gold and silver coins started flowing to North America, these were preferred as a medium of exchange due to their uniformity, allowing for better and more uniform price denomination and giving them higher salability. Further, as more advanced boats and technologies were employed to harvest seashells from the sea, their supply was very highly inflated, leading to a drop in their value and a loss of salability across time. By 1661, seashells stopped being legal tender and eventually lost all monetary role.[4]

This was not just the fate of seashell money in North America; whenever societies employing seashells had access to uniform metal coins, they adopted them and benefited from the switch. Also, the arrival of industrial civilization, with fossil-fuel-powered boats, made scouring the sea for seashells easier, increasing the flow of their production and dropping the stock-to-flow ratio quickly.

Other ancient forms of money include cattle, cherished for their nutritional value, as they were one of the most prized possessions anyone could own and were also salable across space due to their mobility. Cattle continue to play a monetary role today, with many societies using them for payments, especially for dowries. Being bulky and not easily divisible, however, meant cattle were not very useful to solve the problems of divisibility across scales, and so another form of money coexisted along with cattle, and that was salt. Salt was easy to keep for long durations and could be easily divided and grouped into whatever weight was necessary. These historical facts are still apparent in the English language, as the word *pecuniary* is derived from *pecus*, the Latin word for cattle, while the word *salary* is derived from *sal*, the Latin word for salt.[5]

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[4]Ibid: Szabo.

[5]Fekete, Antal. *Whither Gold?* Winner of the 1996 International Currency Prize, Sponsored by Bank Lips, 1997, www.professorfekete.com/articles/AEFWhitherGold.pdf.

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As technology advanced, particularly with metallurgy, humans developed superior forms of money to these artifacts, which began to quickly replace them. These metals proved a better medium of exchange than seashells, stones, beads, cattle, and salt because they could be made into uniform, highly valuable small units that could be moved around far more easily. Another nail in the coffin of artifact money came with the mass utilization of hydrocarbon fuel energy, which increased our productive capacity significantly, allowing for a quick increase in the new supply (flow) of these artifacts, meaning that the forms of money that relied on difficulty of production to protect their high stock-to-flow ratio lost it. With modern hydrocarbon fuels, Rai stones could be quarried easily, aggry beads could be made for very little cost, and seashells could be collected en masse by large boats. As soon as these monies lost their hardness, their holders suffered significant wealth expropriation and the entire fabric of their society fell apart as a result. The Yap Island chiefs who refused O'Keefe's cheap Rai stones understood what most modern economists fail to grasp: a money that is easy to produce is no money at all, and easy money does not make a society richer; on the contrary, it makes it poorer by placing all its hard-earned wealth for sale in exchange for something easy to produce.





Chapter 3

Monetary Metals

As human technical capacity for the production of goods became more sophisticated, and our utilization of metals and commodities grew, many metals began to be produced at large enough quantities and were in large enough demand to make them highly salable and suited for being used as monetary media. These metals' density and relatively high value made moving them around easy, easier than salt or cattle, making them highly salable across space. The production of metals was initially not easy, making it hard to increase their supply quickly and giving them good salability across time.

Due to their durability and physical properties, as well as their relative abundance in earth, some metals were more valuable than others. Iron and copper, because of their relatively high abundance and their susceptibility to corrosion, could be produced in increasing quantities. Existing stockpiles, constantly depleting from corrosion and consumption, could be dwarfed by new production, destroying the value in them. These metals developed a relatively low market value and would be used for smaller transactions. Rarer metals such as silver and gold, on the other hand, were more durable and less likely to corrode or ruin, making

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them more salable across time and useful as a store of value into the future. Gold's virtual indestructibility, in particular, allowed humans to store value across generations, thus allowing us to develop a longer time horizon orientation.

Initially, metals were bought and sold in terms of their weight,[1] but over time, as metallurgy advanced, it became possible to mint them into uniform coins and brand them with their weight, making them far more salable by saving people from having to weigh and assess the metals every time. The three metals most widely used for this role were gold, silver, and copper, and their use as coins was the prime form of money for around 2,500 years, from the time of the Lydian king Croesus, who was the first recorded to have minted gold coins, to the early twentieth century. Gold coins were the goods most salable across time, because they could hold their value over time and resist decay and ruin. They were also the goods most salable across space, because they carried a lot of value in small weights, allowing for easy transportation. Silver coins, on the other hand, had the advantage of being the most salable good across scales, because their lower value per weight unit compared to gold allowed for them to conveniently serve as a medium of exchange for small transactions, while bronze coins would be useful for the least valuable transactions. By standardizing values into easily identifiable units, coins allowed for the creation of large markets, increasing the scope of specialization and trade worldwide. While the best monetary system technologically possible at the time, it still had two major drawbacks: the first was that the existence of two or three metals as the monetary standard created economic problems from the fluctuation of their values over time due to the ebbs of supply and demand, and created problems for owners of these coins, particularly silver, which experienced declines in value due to increases in production and drops in demand. The second, more serious flaw was that governments and counterfeiters could, and frequently did, reduce the precious metal content in these coins, causing their value to decline by transferring a fraction of their purchasing power to the counterfeiters or the government. The reduction in the metal content of the coins compromised the purity and soundness of the money.

[1] Szabo, Nick. "Shelling Out: The Origins of Money." *Satoshi Nakamoto Institute*, 2002, nakamotoinstitute.org/shelling-out/.



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By the nineteenth century, however, with the development of modern banking and the improvement in methods of communication, individuals could transact with paper money and checks backed by gold in the treasuries of their banks and central banks. This made gold-backed transactions possible at any scale, thus obviating the need for silver's monetary role, and gathering all essential monetary salability properties in the gold standard. The gold standard allowed for unprecedented global capital accumulation and trade by uniting the majority of the planet's economy on one sound market-based choice of money. Its tragic flaw, however, was that by centralizing the gold in the vaults of banks, and later central banks, it made it possible for banks and governments to increase the supply of money beyond the quantity of gold they held, devaluing the money and transferring part of its value from the money's legitimate holders to the governments and banks.

Why Gold?

To understand how commodity money emerges, we return in more detail to the easy money trap we first introduced in Chapter 1, and begin by differentiating between a good's *market demand* (demand for consuming or holding the good for its own sake) and its *monetary demand* (demand for a good as a medium of exchange and store of value). Any time a person chooses a good as a store of value, she is effectively increasing the demand for it beyond the regular market demand, which will cause its price to rise. For example, market demand for copper in its various industrial uses is around 20 million tons per year, at a price of around $5,000 per ton, and a total market valued around $100 billion. Imagine a billionaire deciding he would like to store $10 billion of his wealth in copper. As his bankers run around trying to buy 10% of annual global copper production, they would inevitably cause the price of copper to increase. Initially, this sounds like a vindication of the billionaire's monetary strategy: the asset he decided to buy has already appreciated before he has even completed his purchase. Surely, he reasons, this appreciation will cause more people to buy more copper as a store of value, bringing the price up even more.

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But even if more people join him in monetizing copper, our hypothetical copper-obsessed billionaire is in trouble. The rising price makes copper a lucrative business for workers and capital across the world. The quantity of copper under the earth is beyond our ability to even measure, let alone extract through mining, so practically speaking, the only binding restraint on how much copper can be produced is how much labor and capital is dedicated to the job. More copper can always be made with a higher price, and the price and quantity will continue to rise until they satisfy the monetary investors' demand; let's assume that happens at 10 million extra tons and $10,000 per ton. At some point, monetary demand must subside, and some holders of copper will want to offload some of their stockpiles to purchase other goods, because, after all, that was the point of buying copper.

After the monetary demand subsides, all else being equal, the copper market would go back to its original supply-and-demand conditions, with 20 million annual tons selling for $5,000 each. But as the holders begin to sell their accumulated stocks of copper, the price will drop significantly below that. The billionaire will have lost money in this process; as he was driving the price up, he bought most of his stock for more than $5,000 a ton, but now his entire stock is valued below $5,000 a ton. The others who joined him later bought at even higher prices and will have lost even more money than the billionaire himself.

This model is applicable for all consumable commodities such as copper, zinc, nickel, brass, or oil, which are primarily consumed and destroyed, not stockpiled. Global stockpiles of these commodities at any moment in time are around the same order of magnitude as new annual production. New supply is constantly being generated to be consumed. Should savers decide to store their wealth in one of these commodities, their wealth will only buy a fraction of global supply before bidding the price up enough to absorb all their investment, because they are competing with the consumers of this commodity who use it productively in industry. As the revenue to the producers of the good increases, they can then invest in increasing their production, bringing the price crashing down again, robbing the savers of their wealth. The net effect of this entire episode is the transfer of the wealth of the misguided savers to the producers of the commodity they purchased.



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This is the anatomy of a market bubble: increased demand causes a sharp rise in prices, which drives further demand, raising prices further, incentivizing increased production and increased supply, which inevitably brings prices down, punishing everyone who bought at a price higher than the usual market price. Investors in the bubble are fleeced while producers of the asset benefit. For copper and almost every other commodity in the world, this dynamic has held true for most of recorded history, consistently punishing those who choose these commodities as money by devaluing their wealth and impoverishing them in the long run, and returning the commodity to its natural role as a market good, and not a medium of exchange.

For anything to function as a good store of value, it has to beat this trap: it has to appreciate when people demand it as a store of value, but its producers have to be constrained from inflating the supply significantly enough to bring the price down. Such an asset will reward those who choose it as their store of value, increasing their wealth in the long run as it becomes the prime store of value, because those who chose other commodities will either reverse course by copying the choice of their more successful peers, or will simply lose their wealth.

The clear winner in this race throughout human history has been gold, which maintains its monetary role due to two unique physical characteristics that differentiate it from other commodities: first, gold is so chemically stable that it is virtually impossible to destroy, and second, gold is impossible to synthesize from other materials (alchemists' claims notwithstanding) and can only be extracted from its unrefined ore, which is extremely rare in our planet.

The chemical stability of gold implies that virtually all of the gold ever mined by humans is still more or less owned by people around the world. Humanity has been accumulating an ever-growing hoard of gold in jewelry, coins, and bars, which is never consumed and never rusts or disintegrates. The impossibility of synthesizing gold from other chemicals means that the only way to increase the supply of gold is by mining gold from the earth, an expensive, toxic, and uncertain process in which humans have been engaged for thousands of years with ever-diminishing returns. This all means that the existing stockpile of gold held by people around the world is the product of thousands of years of gold production, and is orders of magnitude larger than new annual production. Over the

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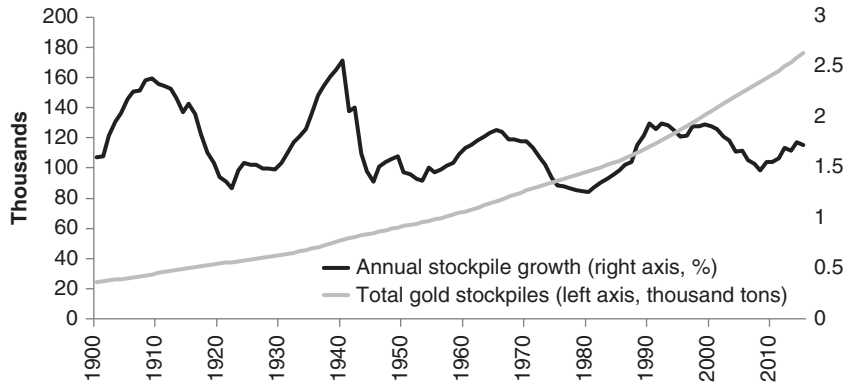


Figure 1 Global gold stockpiles and annual stockpile growth rate.

past seven decades with relatively reliable statistics, this growth rate has always been around 1.5%, never exceeding 2%. (See Figure 1.[2])

To understand the difference between gold and any consumable commodity, imagine the effect of a large increase in demand for it as a store of value that causes the price to spike and annual production to double. For any consumable commodity, this doubling of output will dwarf any existing stockpiles, bringing the price crashing down and hurting the holders. For gold, a price spike that causes a doubling of annual production will be insignificant, increasing stockpiles by 3% rather than 1.5%. If the new increased pace of production is maintained, the stockpiles grow faster, making new increases less significant. It remains practically impossible for goldminers to mine quantities of gold large enough to depress the price significantly.

Only silver comes close to gold in this regard, with an annual supply growth rate historically around 5–10%, rising to around 20% in the modern day. This is higher than that of gold for two reasons: First, silver does corrode and can be consumed in industrial processes, which means the existing stockpiles are not as large relative to annual production as gold's stockpiles are relative to its annual production. Second, silver is less rare than gold in the crust of the earth and easier to refine. Because of having the second highest stock-to-flow ratio, and its lower value per

[2] *Source*: US Geological Survey. Global Gold Stockpiles and Annual Stockpile Growth Rate.

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unit of weight than gold, silver served for millennia as the main money used for smaller transactions, complementing gold, whose high value meant dividing it into very small units was impractical. The adoption of the international gold standard allowed for payments in paper backed by gold at any scale, as will be discussed in more detail later in this chapter, which obviated silver's monetary role. With silver no longer required for smaller transactions, it soon lost its monetary role and became an industrial metal, losing value compared to gold. Silver may maintain its sporting connotation for second place, but as nineteenth-century technology made payments possible without having to move the monetary unit itself, second place in monetary competition was equivalent to losing out.

This explains why the silver bubble has popped before and will pop again if it ever inflates: as soon as significant monetary investment flows into silver, it is not as difficult for producers to increase the supply significantly and bring the price crashing down, taking the savers' wealth in the process. The best-known example of the easy-money trap comes from silver itself, of all commodities. Back in the late 1970s, the very affluent Hunt brothers decided to bring about the remonetization of silver and started buying enormous quantities of silver, driving the price up. Their rationale was that as the price rose, more people would want to buy, which would keep the price rising, which in turn would lead to people wanting to be paid in silver. Yet, no matter how much the Hunt brothers bought, their wealth was no match for the ability of miners and holders of silver to keep selling silver onto the market. The price of silver eventually crashed and the Hunt brothers lost over $1 billion, probably the highest price ever paid for learning the importance of the stock-to-flow ratio, and why not all that glitters is gold.[3] (See Figure 2.[4])

It is this consistently low rate of supply of gold that is the fundamental reason it has maintained its monetary role throughout human history, a role it continues to hold today as central banks continue to hold significant supplies of gold to protect their paper currencies. Official central bank reserves are at around 33,000 tons, or a sixth of total

[3] Rudolph, Barbara. "Big Bill for a Bullion Binge." *TIME*, 29 Aug. 1988.

[4] *Source*: U.S. Geological Survey data for gold. Silver Institute data for silver, BP.com statistical review for oil. Author's estimates from various media sources for copper.



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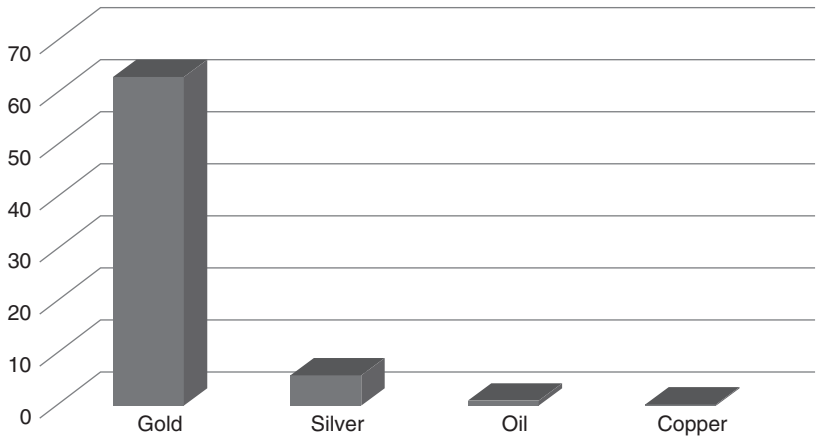


Figure 2 Existing stockpiles as a multiple of annual production.

above-ground gold. The high stock-to-flow ratio of gold makes it the commodity with the lowest *price elasticity of supply*, which is defined as the percentage increase in quantity supplied over the percentage increase in price. Given that the existing supply of gold held by people everywhere is the product of thousands of years of production, an *X*% increase in price may cause an increase in new mining production, but that increase will be trivial compared to existing stockpiles. For instance, the year 2006 witnessed a 36% rise in the spot price of gold. For any other commodity, this would be expected to increase mining output significantly to flood markets and bring the price down. Instead, annual production in 2006 was 2,370 tons, 100 tons less than in 2005, and it would drop a further 10 tons in 2007. Whereas the new supply was 1.67% of existing stockpiles in 2005, it was 1.58% of existing stockpiles in 2006, and 1.54% of existing stockpiles in 2007. Even a 35% rise in price can lead to no appreciable increase in the supply of new gold onto the market. According to the U.S. Geological Survey, the single biggest annual increase in production was around 15% in the year 1923, which translated to an increase in stockpiles around only 1.5%. Even if production were to double, the likely increase in stockpiles would only be around 3–4%. The highest annual increase in global stockpiles happened in 1940, when stockpiles rose by around 2.6%. Not once has the annual stockpile growth exceeded that number, and not once since 1942 has it exceeded 2%.

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As the production of metals began to proliferate, ancient civilizations in China, India, and Egypt began to use copper, and later silver, as money, as these two were relatively hard to manufacture at the time and allowed for good salability across time and space. Gold was highly prized in these civilizations, but its rarity meant its salability for transactions was limited. It was in Lydia, in modern day Turkey, where gold was first minted into regular coins for trade, under King Croesus. This invigorated global trade as gold's global appeal saw the coin spread far and wide. Since then, the turns of human history have been closely intertwined with the soundness of money. Human civilization flourished in times and places where sound money was widely adopted, while unsound money all too frequently coincided with civilizational decline and societal collapse.

Roman Golden Age and Decline

The denarius was the silver coin that traded at the time of the Roman Republic, containing 3.9 grams of silver, while gold became the most valuable money in the civilized areas of the world at the time and gold coins were becoming more widespread. Julius Caesar, the last dictator of the Roman Republic, created the aureus coin, which contained around 8 grams of gold and was widely accepted across Europe and the Mediterranean, increasing the scope of trade and specialization in the Old World. Economic stability reigned for seventy-five years, even through the political upheaval of his assassination, which saw the Republic transformed into an Empire under his chosen successor, Augustus. This continued until the reign of the infamous emperor Nero, who was the first to engage in the Roman habit of "coin clipping," wherein the Emperor would collect the coins of the population and mint them into newer coins with less gold or silver content.

For as long as Rome could conquer new lands with significant wealth, its soldiers and emperors could enjoy spending their loot, and emperors even decided to buy themselves popularity by mandating artificially low prices of grains and other staples, sometimes even granting them for free. Instead of working for a living in the countryside, many peasants would leave their farms to move to Rome,

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where they could live better lives for free. With time, the Old World no longer had prosperous lands to be conquered, the ever-increasing lavish lifestyle and growing military required some new source of financing, and the number of unproductive citizens living off the emperor's largesse and price controls increased. Nero, who ruled from 54–68 AD, had found the formula to solve this, which was highly similar to Keynes's solution to Britain's and the U.S.'s problems after World War I: devaluing the currency would at once reduce the real wages of workers, reduce the burden of the government in subsidizing staples, and provide increased money for financing other government expenditure.

The aureus coin was reduced from 8 to 7.2 grams, while the denarius's silver content was reduced from 3.9 to 3.41g. This provided some temporary relief, but had set in motion the highly destructive self-reinforcing cycle of popular anger, price controls, coin debasement, and price rises, following one another with the predictable regularity of the four seasons.[5]

Under the reign of Caracalla (AD 211–217), the gold content was further reduced to 6.5 grams, and under Diocletian (AD 284–305) it was further reduced to 5.5g, before he introduced a replacement coin called the solidus, with only 4.5 grams of gold. On Diocletian's watch, the denarius only had traces of silver to cover its bronze core, and the silver would disappear quite quickly with wear and tear, ending the denarius as a silver coin. As inflationism intensified in the third and fourth centuries, with it came the misguided attempts of the emperors to hide their inflation by placing price controls on basic goods. As market forces sought to adjust prices upward in response to the debasement of the currency, price ceilings prevented these price adjustments, making it unprofitable for producers to engage in production. Economic production would come to a standstill until a new edict allowed for the liberalization of prices upward.

With this fall in the value of its money, the long process of terminal decline of the empire resulted in a cycle that might appear familiar to modern readers: coin clipping reduced the aureus's real value, increasing

[5]See Robert Schuettinger and Eamonn Butler's highly entertaining *Forty Centuries of Price and Wage Controls: How Not to Fight Inflation*.

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the money supply, allowing the emperor to continue imprudent overspending, but eventually resulting in inflation and economic crises, which the misguided emperors would attempt to ameliorate via further coin clipping. Ferdinand Lips summarizes this process with a lesson to modern readers:

> It should be of interest to modern Keynesian economists, as well as to the present generation of investors, that although the emperors of Rome frantically tried to "manage" their economies, they only succeeded in making matters worse. Price and wage controls and legal tender laws were passed, but it was like trying to hold back the tides. Rioting, corruption, lawlessness and a mindless mania for speculation and gambling engulfed the empire like a plague. With money so unreliable and debased, speculation in commodities became far more attractive than producing them.[6]

The long-term consequences for the Roman Empire were devastating. Although Rome up until the second century AD may not be characterized as a full-fledged free market capitalist economy, because it still had plenty of government restraints on economic activity, with the aureus it nonetheless established what was then the largest market in human history with the largest and most productive division of labor the world had ever known.[7] Citizens of Rome and the major cities obtained their basic necessities by trade with the far-flung corners of the empire, and this helps explain the growth in prosperity, and the devastating collapse the empire suffered when this division of labor fell apart. As taxes increased and inflation made price controls unworkable, the urbanites of the cities started fleeing to empty plots of land where they could at least have a chance of living in self-sufficiency, where their lack of income spared them having to pay taxes. The intricate civilizational edifice of the Roman Empire and the large division of labor across Europe and the Mediterranean began to crumble, and its descendants became self-sufficient peasants scattered in isolation and would soon turn into serfs living under feudal lords.

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[6]Lips, Ferdinand. *Gold Wars: The Battle Against Sound Money as Seen from a Swiss Perspective*. New York, Foundation for the Advancement of Monetary Education, 2001.

[7]Mises, Ludwig von. *Human Action: The Scholar's Edition*. Auburn, AL, Ludwig von Mises Institute, 1998.

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Byzantium and the Bezant

The emperor Diocletian has forever had his name associated with fiscal and monetary chicanery, and the Empire reached a nadir under his rule. A year after he abdicated, however, Constantine the Great took over the reins of the empire and reversed its fortunes by adopting economically responsible polices and reforms. Constantine, who was the first Christian emperor, committed to maintaining the solidus at 4.5 grams of gold without clipping or debasement and started minting it in large quantities in 312 AD. He moved east and established Constantinople at the meeting point of Asia and Europe, birthing the Eastern Roman Empire, which took the solidus as its coin. While Rome continued its economic, social, and cultural deterioration, finally collapsing in 476 AD, Byzantium survived for 1,123 years while the solidus became the longest-serving sound currency in human history.

The legacy of Constantine in maintaining the integrity of the solidus made it the world's most recognizable and widely accepted currency, and it came to be known as the *bezant*. While Rome burned under bankrupt emperors who could no longer afford to pay their soldiers as their currencies collapsed, Constantinople thrived and prospered for many more centuries with fiscal and monetary responsibility. While the Vandals and the Visigoths ran rampage in Rome, Constantinople remained prosperous and free from invasion for centuries. As with Rome, the fall of Constantinople happened only after its rulers had started devaluing the currency, a process that historians believe began in the reign of Constantine IX Monomachos (1042–1055).[8] Along with monetary decline came the fiscal, military, cultural, and spiritual decline of the Empire, as it trudged on with increasing crises until it was overtaken by the Ottomans in 1453.

Even after it was debased and its empire fell, the bezant lived on by inspiring another form of sound money that continues to circulate widely to this day in spite of not being the official currency of any nation anymore, and that is the Islamic dinar. As Islam rose during the golden age of Byzantium, the bezant and coins similar to it in weight and size

[8]Luscombe, David, and Jonathan Riley-Smith. *The New Cambridge Medieval History: Volume 4, C.1024–1198*. Cambridge University Press, 2008, p. 255.



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were circulating in the regions to which Islam had spread. The Umayyad Caliph Abdul-Malik ibn Marwan defined the weight and value of the Islamic dinar and imprinted it with the Islamic *shahada* creed in 697 AD. The Umayyad dynasty fell, and after it several other Islamic states, and yet the dinar continues to be held and to circulate widely in Islamic regions in the original weight and size specifications of the bezant, and is used in dowries, gifts, and various religious and traditional customs to this day. Unlike the Romans and the Byzantines, Arab and Muslim civilizations' collapse was not linked to the collapse of their money as they maintained the integrity of their currencies for centuries. The solidus, first minted by Diocletian in AD 301, has changed its name to the bezant and the Islamic dinar, but it continues to circulate today. Seventeen centuries of people the world over have used this coin for transactions, emphasizing the salability of gold across time.

The Renaissance

After the economic and military collapse of the Roman Empire, feudalism emerged as the prime mode of organizing society. The destruction of sound money was pivotal in turning the former citizens of the Roman Empire into serfs under the mercy of their local feudal lords. Gold was concentrated in the hands of the feudal lords, and the main forms of money available for the peasantry of Europe at the time were copper and bronze coins, whose supply was easy to inflate as industrial production of these metals continued to become easier with the advance of metallurgy, making them terrible stores of value, as well as silver coins that were usually debased, cheated, and nonstandardized across the continent, giving them poor salability across space and limiting the scope of trade across the continent.

Taxation and inflation had destroyed the wealth and savings of the people of Europe. New generations of Europeans came to the world with no accumulated wealth passed on from their elders, and the absence of a widely accepted sound monetary standard severely restricted the scope for trade, closing societies off from one another and enhancing parochialism as once-prosperous and civilized trading societies fell into the Dark Ages of serfdom, diseases, closed-mindedness, and religious persecution.

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While it is widely recognized that the rise of the city-states dragged Europe out of the Dark Ages and into the Renaissance, the role of sound money in this rise is less recognized. It was in the city-states that humans could live with the freedom to work, produce, trade, and flourish, and that was to a large extent the result of these city-states adopting a sound monetary standard. It all began in Florence in 1252, when the city minted the florin, the first major European sound coinage since Julius Caesar's aureus. Florence's rise made it the commercial center of Europe, with its florin becoming the prime European medium of exchange, allowing its banks to flourish across the entire continent. Venice was the first to follow Florence's example with its minting of the ducat, of the same specifications as the florin, in 1270, and by the end of the fourteenth century more than 150 European cities and states had minted coins of the same specifications as the florin, allowing their citizens the dignity and freedom to accumulate wealth and trade with a sound money that was highly salable across time and space, and divided into small coins, allowing for easy divisibility. With the economic liberation of the European peasantry came the political, scientific, intellectual, and cultural flourishing of the Italian city-states, which later spread across the European continent. Whether in Rome, Constantinople, Florence, or Venice, history shows that a sound monetary standard is a necessary prerequisite for human flourishing, without which society stands on the precipice of barbarism and destruction.

Although the period following the introduction of the florin witnessed an improvement in the soundness of money, with more and more Europeans able to adopt gold and silver for saving and trade, and the extent of markets expanding across Europe and the world, the situation was far from perfect. There were still many periods during which various sovereigns would debase their people's currency to finance war or lavish expenditure. Given that they were used physically, silver and gold complemented each other: gold's high stock-to-flow ratio meant it was ideal as a long-term store of value and a means of large payments, but silver's lower value per unit of weight made it easily divisible into quantities suitable for smaller transactions and for being held for shorter durations. While this arrangement had benefits, it had one major drawback: the fluctuating rate of exchange between gold and silver created trade and calculation problems. Attempts to fix the price of the two

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currencies relative to one another were continuously self-defeating, but gold's monetary edge was to win out.

As sovereigns set an exchange rate between the two commodities, they would change holders' incentives to hold or spend them. This inconvenient bimetallism continued for centuries across Europe and the world, but as with the move from salt, cattle, and seashells to metals, the inexorable advance of technology was to provide a solution to it.

Two particular technological advancements would move Europe and the world away from physical coins and in turn help bring about the demise of silver's monetary role: the telegraph, first deployed commercially in 1837, and the growing network of trains, allowing transportation across Europe. With these two innovations, it became increasingly feasible for banks to communicate with each other, sending payments efficiently across space when needed and debiting accounts instead of having to send physical payments. This led to the increased use of bills, checks, and paper receipts as monetary media instead of physical gold and silver coins.

More nations began to switch to a monetary standard of paper fully backed by, and instantly redeemable into, precious metals held in vaults. Some nations would choose gold, and others would choose silver, in a fateful decision that was to have enormous consequences. Britain was the first to adopt a modern gold standard in 1717, under the direction of physicist Isaac Newton, who was the warden of the Royal Mint, and the gold standard would play a great role in Britain advancing its trade across its empire worldwide. Britain would remain under a gold standard until 1914, although it would suspend it during the Napoleonic wars from 1797 to 1821. The economic supremacy of Britain was intricately linked to its being on a superior monetary standard, and other European countries began to follow it. The end of the Napoleonic wars heralded the beginning of the golden age of Europe, as, one by one, the major European nations began adopting the gold standard. The more nations officially adopted the gold standard, the more marketable gold became and the larger the incentive became for other nations to join.

Further, instead of individuals having to carry gold and silver coins for large and small transactions, respectively, they could now store their wealth in gold in banks while using paper receipts, bills, and checks to make payments of any size. The holders of paper receipts could just use

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them to make payment themselves; bills were discounted by banks and used for clearance and checks could be cashed from the banks that issued them. This solved the problem of gold's salability across scales, making gold the best monetary medium—for as long as the banks hoarding people's gold would not increase the supply of papers they issued as receipts.

With these media being backed by physical gold in the vaults and allowing payment in whichever quantity or size, there was no longer a real need for silver's role in small payments. The death knell for silver's monetary role was the end of the Franco-Prussian war, when Germany extracted an indemnity of £200 million in gold from France and used it to switch to a gold standard. With Germany now joining Britain, France, Holland, Switzerland, Belgium, and others on a gold standard, the monetary pendulum had swung decisively in favor of gold, leading to individuals and nations worldwide who used silver to witness a progressive loss of their purchasing power and a stronger incentive to shift to gold. India finally switched from silver to gold in 1898, while China and Hong Kong were the last economies in the world to abandon the silver standard in 1935.

For as long as gold and silver were used for payment directly, they both had a monetary role to play and their price relative to one another remained largely constant across time, at a ratio between twelve and fifteen ounces of silver per ounce of gold, in the same range as their relative scarcity in the crust of the earth and the relative difficulty and cost of extracting them. But as paper and financial instruments backed by these metals became more and more popular, there was no more justification for silver's monetary role, and individuals and nations shifted to holding gold, leading to a significant collapse in the price of silver, from which it would not recover. The average ratio between the two over the twentieth century was 47:1, and in 2017, it stood at 75:1. While gold still has a monetary role to play, as evidenced by central banks' hoarding of it, silver has arguably lost its monetary role. (See Figure 3.[9])

The demonetization of silver had a significantly negative effect on the nations that were using it as a monetary standard at the time. India witnessed a continuous devaluation of its rupee compared to gold-based

[9] *Source*: Officer, Lawrence H., and Samuel H. Williamson. *The Price of Gold, 1257–Present*. MeasuringWorth, 2017, *www.measuringworth.com/gold/*.

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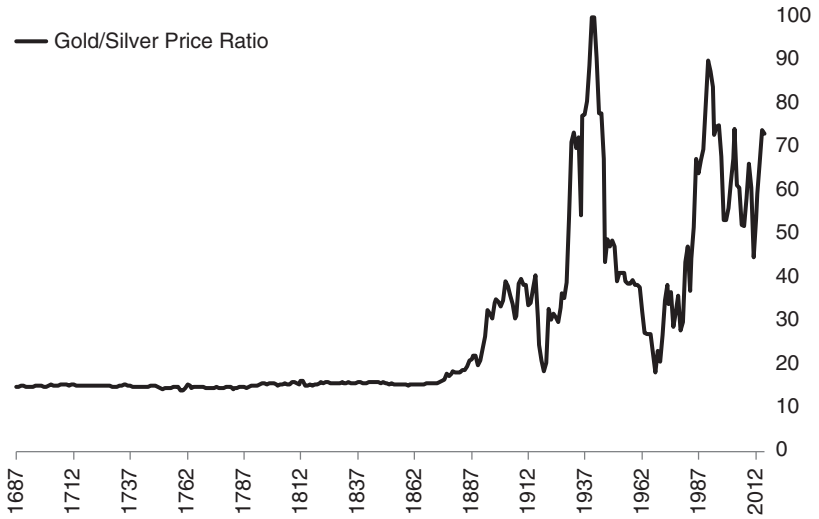


Figure 3 Price of gold in silver ounces, 1687–2017.

European countries, which led the British colonial government to increase taxes to finance its operation, leading to growing unrest and resentment of British colonialism. By the time India shifted the backing of its rupee to the gold-backed pound sterling in 1898, the silver backing its rupee had lost 56% of its value in the twenty-seven years since the end of the Franco-Prussian War. For China, which stayed on the silver standard until 1935, its silver (in various names and forms) lost 78% of its value over the period. It is the author's opinion that the history of China and India, and their failure to catch up to the West during the twentieth century, is inextricably linked to this massive destruction of wealth and capital brought about by the demonetization of the monetary metal these countries utilized. The demonetization of silver in effect left the Chinese and Indians in a situation similar to west Africans holding aggri beads as Europeans arrived: domestic hard money was easy money for foreigners, and was being driven out by foreign hard money, which allowed foreigners to control and own increasing quantities of the capital and resources of China and India during the period. This is a historical lesson of immense significance, and should be kept in mind by anyone who thinks his refusal of bitcoin means he doesn't have to

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deal with it. History shows it is not possible to insulate yourself from the consequences of others holding money that is harder than yours.

With gold in the hands of increasingly centralized banks, it gained salability across time, scales, and location, but lost its property as cash money, making payments in it subject to the agreement of the financial and political authorities issuing receipts, clearing checks, and hoarding the gold. Tragically, the only way gold was able to solve the problems of salability across scales, space, and time was by being centralized and thus falling prey to the major problem of sound money emphasized by the economists of the twentieth century: individual sovereignty over money and its resistance to government centralized control. We can thus understand why nineteenth-century sound money economists like Menger focused their understanding of money's soundness on its salability as a market good, whereas twentieth-century sound money economists, like Mises, Hayek, Rothbard, and Salerno, focused their analysis of money's soundness on its resistance to control by a sovereign. Because the Achilles heel of twentieth century money was its centralization in the hands of the government, we will see later how the money invented in the twenty-first century, bitcoin, was designed primarily to avoid centralized control.

La Belle Époque

The end of the Franco-Prussian War in 1871, and the consequent shift of all major European powers onto the same monetary standard, namely gold, led to a period of prosperity and flourishing that continues to appear more amazing with time and in retrospect. A case can be made for the nineteenth century—in particular, the second half of it—being the greatest period for human flourishing, innovation, and achievement that the world had ever witnessed, and the monetary role of gold was pivotal to it. With silver and other media of exchange increasingly demonetized, the majority of the planet used the same golden monetary standard, allowing the improvements in telecommunications and transportation to foster global capital accumulation and trade like never before.

Different currencies were simply different weights of physical gold, and the exchange rate between one nation's currency and the other

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Table 1 Major European Economies' Periods Under the Gold Standard

Currency	Period Under Gold Standard	Years
French Franc	1814–1914	100 years
Dutch Guilder	1816–1914	98 years
Pound Sterling	1821–1914	93 years
Swiss Franc	1850–1936	86 years
Belgian Franc	1832–1914	82 years
Swedish Krona	1873–1931	58 years
German Mark	1875–1914	39 years
Italian Lira	1883–1914	31 years

was the simple conversion between different weight units, as straightforward as converting inches to centimeters. The British pound was defined as 7.3 grams of gold, while the French franc was 0.29 grams of gold and the Deutschmark 0.36 grams, meaning the exchange rate between them was necessarily fixed at 25.2 French francs and 20.4 Deutschmark per pound. In the same way metric and imperial units are just a way to measure the underlying length, national currencies were just a way to measure economic value as represented in the universal store of value, gold. Some countries' gold coins were fairly salable in other countries, as they were just gold. Each country's money supply was not a metric to be determined by central planning committees stocked with Ph.D. holders, but the natural working of the market system. People held as much money as they pleased and spent as much as they desired on local or foreign production, and the actual money supply was not even easily measurable.

The soundness of money was reflected in free trade across the world, but perhaps more importantly, was increasing savings rates across most advanced societies that were on the gold standard, allowing for capital accumulation to finance industrialization, urbanization, and the technological improvements that have shaped our modern life. (See Table 1.[10])

By 1900, around fifty nations were officially on the gold standard, including all industrialized nations, while the nations that were not on an official gold standard still had gold coins being used as the main medium

[10] *Source*: Lips, Ferdinand. *Gold Wars: The Battle Against Sound Money as Seen from a Swiss Perspective*. New York, Foundation for the Advancement of Monetary Education, 2001.

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of exchange. Some of the most important technological, medical, economic, and artistic human achievements were invented during the era of the gold standard, which partly explains why it was known as *la belle époque*, or the beautiful era, across Europe. Britain witnessed the peak years of Pax Britannica, where the British Empire expanded worldwide and was not engaged in large military conflicts. In 1899, when American writer Nellie Bly set out on her record-breaking journey around the world in seventy-two days, she carried British gold coins and Bank of England notes with her.[11] It was possible to circumnavigate the globe and use one form of money everywhere Nellie went.

In the United States this era was called the Gilded Age, where economic growth boomed after the restoration of the gold standard in 1879 in the wake of the American Civil War. It was only interrupted by one episode of monetary insanity, which was effectively the last dying pang of silver as money, discussed in Chapter 6, when the Treasury tried to remonetize silver by mandating it as money. This caused a large increase in the money supply and a bank run by those seeking to sell Treasury notes and silver to gold. The result was the recession of 1893, after which U.S. economic growth resumed.

With the majority of the world on one sound monetary unit, there was never a period that witnessed as much capital accumulation, global trade, restraint on government, and transformation of living standards worldwide. Not only were the economies of the west far freer back then, the societies themselves were far freer. Governments had very few bureaucracies focused on micromanaging the lives of citizens. As Mises described it:

> The gold standard was the world standard of the age of capitalism, increasing welfare, liberty, and democracy, both political and economic. In the eyes of the free traders its main eminence was precisely the fact that it was an international standard as required by international trade and the transactions of the international money and capital markets. It was the medium of exchange by means of which Western industrialism and Western capital had borne Western civilization to the remotest parts of the earth's surface, everywhere destroying the fetters of old-aged prejudices and superstitions, sowing

[11] Bly, Nellie. *Around the World in Seventy-Two Days*. New York, Pictorial Weeklies, 1890.



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> the seeds of new life and new well-being, freeing minds and souls, and creating riches unheard of before. It accompanied the triumphal unprecedented progress of Western liberalism ready to unite all nations into a community of free nations peacefully cooperating with one another.
>
> It is easy to understand why people viewed the gold standard as the symbol of this greatest and most beneficial of all historical changes.[12]

This world came crashing down in the catastrophic year 1914, which was not only the year of the outbreak of World War I, but the year that the world's major economies went off of the gold standard and replaced it with unsound government money. Only Switzerland and Sweden, who remained neutral during World War I, were to remain on a gold standard into the 1930s. The era of government-controlled money was to commence globally after that, with unmitigated disastrous consequences.

While the gold standard of the nineteenth century was arguably the closest thing that the world had ever seen to an ideal sound money, it nonetheless had its flaws. First, governments and banks were always creating media of exchange beyond the quantity of gold in their reserves. Second, many countries used not just gold in their reserves, but also currencies of other countries. Britain, as the global superpower at that time, had benefited from having its money used as a reserve currency all around the world, resulting in its reserves of gold being a tiny fraction of its outstanding money supply. With growing international trade relying on settlement of large quantities of money across the world, the Bank of England's banknotes became, in the minds of many at the time, "as good as gold." While gold was very hard money, the instruments used for settlements of payments between central banks, although nominally redeemable in gold, ended up in practice being easier to produce than gold.

These two flaws meant that the gold standard was always vulnerable to a run on gold in any country where circumstances might lead a large enough percentage of the population to demand redemption of their paper money in gold. The fatal flaw of the gold standard at the heart of these two problems was that settlement in physical gold is

[12]Mises, Ludwig von. *Human Action: The Scholar's Edition.* pp. 472–3.



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cumbersome, expensive, and insecure, which meant it had to rely on centralizing physical gold reserves in a few locations—banks and central banks—leaving them vulnerable to being taken over by governments. As the number of payments and settlements conducted in physical gold became an infinitely smaller fraction of all payments, the banks and central banks holding the gold could create money unbacked by physical gold and use it for settlement. The network of settlement became valuable enough that its owners' credit was effectively monetized. As the ability to run a bank started to imply money creation, governments naturally gravitated to taking over the banking sector through central banking. The temptation was always too strong, and the virtually infinite financial wealth this secured could not only silence dissent, but also finance propagandists to promote such ideas. Gold offered no mechanism for restraining the sovereigns, and had to rely on trust in them not abusing the gold standard and the population remaining eternally vigilant against them doing so. This might have been feasible when the population was highly educated and knowledgeable about the dangers of unsound money, but with every passing generation displaying the intellectual complacence that tends to accompany wealth,[13] the siren song of con artists and court-jester economists would prove increasingly irresistible for more of the population, leaving only a minority of knowledgeable economists and historians fighting an uphill battle to convince people that wealth can't be generated by tampering with the money supply, that allowing a sovereign the control of the money can only lead to them increasing their control of everyone's life, and that civilized human living itself rests on the integrity of money providing a solid foundation for trade and capital accumulation.

Gold being centralized made it vulnerable to having its monetary role usurped by its enemies, and gold simply had too many enemies, as Mises himself well understood:

> The nationalists are fighting the gold standard because they want to sever their countries from the world market and to establish national autarky as far as possible. Interventionist governments and pressure groups are fighting the gold standard because they consider it the

[13] Glubb, John. *The Fate of Empires and Search for Survival*. Edinburgh, Scotland, William Blackwood and Sons, 1978.

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> most serious obstacle to their endeavours to manipulate prices and wage rates. But the most fanatical attacks against gold are made by those intent upon credit expansion. With them credit expansion is the panacea for all economic ills.[14]
>
> The gold standard removes the determination of cash-induced changes in purchasing power from the political arena. Its general acceptance requires the acknowledgement of the truth that one cannot make all people richer by printing money. The abhorrence of the gold standard is inspired by the superstition that omnipotent governments can create wealth out of little scraps of paper [. . .] The governments were eager to destroy it, because they were committed to the fallacies that credit expansion is an appropriate means of lowering the rate of interest and of "improving" the balance of trade [. . .] People fight the gold standard because they want to substitute national autarky for free trade, war for peace, totalitarian government omnipotence for liberty.[15]

The twentieth century began with governments bringing their citizens' gold under their control through the invention of the modern central bank on the gold standard. As World War I started, the centralization of these reserves allowed these governments to expand the money supply beyond their gold reserves, reducing the value of their currency. Yet central banks continued to confiscate and accumulate more gold until the 1960s, where the move toward a U.S. dollar global standard began to shape up. Although gold was supposedly demonetized fully in 1971, central banks continued to hold significant gold reserves, and only disposed of them slowly, before returning to buying gold in the last decade. Even as central banks repeatedly declared the end of gold's monetary role, their actions in maintaining their gold reserves ring truer. From a monetary competition perspective, keeping gold reserves is a perfectly rational decision. Keeping reserves in foreign governments' easy money only will cause the value of the country's currency to devalue along with the reserve currencies, while the seigniorage accrues to the issuer of the reserve currency, not the nation's central bank. Further, should central banks sell all their gold holdings (estimated at around 20% of global gold stockpiles), the most likely impact is that

[14] Mises, Ludwig von. *Human Action: The Scholar's Edition*. p. 473.
[15] Ibid., p. 474.

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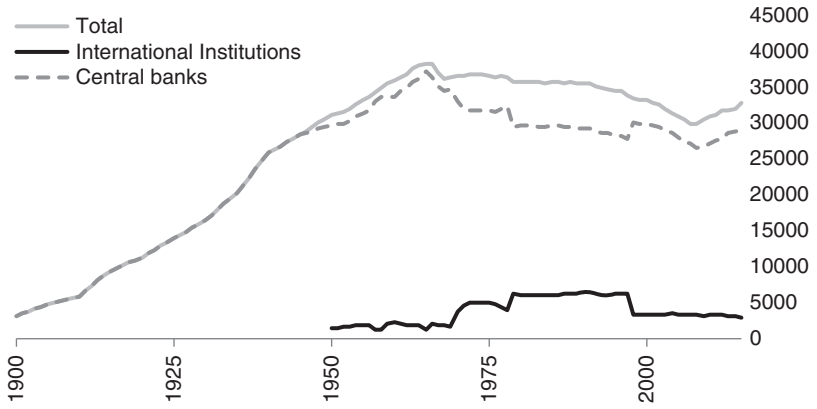


Figure 4 Central bank official gold reserves, tons.

gold, being highly prized for its industrial and aesthetic uses, would be bought up very quickly with little depreciation of its price and the central banks would be left without any gold reserves. The monetary competition between easy government money and hard gold will likely result in one winner in the long-run. Even in a world of government money, governments have not been able to decree gold's monetary role away, as their actions speak louder than their words. (See Figure 4.[16])

[16] *Source*: World Gold Council. *Reserve Statistics*, www.gold.org/goldhub/data/monthly-central-bank-statistics.